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Geographical, Ecological, and Genetic Drivers of Gut Microbial Diversity in Native and Invasive Minnows (Leuciscidae: *Cyprinella*)

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ABSTRACT

The gut microbiome is important for many physiological processes that are critical in the adaptation of an animal to its environment. Conversely, abrupt ecological changes, as in the colonisation of a new territory, may also influence the microbiome. Therefore, anthropogenic introductions of invasive species offer a natural model in which to study these relationships. We compared the gut microbiomes (16S rRNA gene) of four freshwater fish species of the genus *Cyprinella*, including both native and introduced populations of the prolific invader *C. lutrensis*, to investigate if differences in their diversity and structure are determined by their host or depend more on the ecology and geographical location where they occur. Our results suggest that at this taxonomic level, the external environment of the fish is the strongest corollary of microbial diversity and community composition of the gut, followed to a lesser extent by species identity and genetic factors. Our findings emphasise the dynamic nature of the minnow gut microbiome, with high individual variation and rapid changes over time. We also found that new invasions may reduce the invader's gut microbiome alpha diversity while not conferring any clear distinction compared with cohabiting native species. This research addresses the perennial question of whether nature or nurture plays a greater role in shaping the gut microbiome, revealing the intricate interplay of factors and scales involved.

1 | Introduction

Animal digestive systems are home to a vast array of microorganisms, including bacteria, archaea, fungi and other microbes collectively known as gut microbiota that together with the surrounding environment constitute the microbiome (Marchesi and Ravel 2015). In the last decade, the advent of massive parallel sequencing technologies has led to a tremendous amount of discovery and has opened questions regarding the role that a microbiome plays regarding its host organism (e.g., physiological processes like digestion, immune response or metabolism), and how that microbiome is acquired (Berg et al. 2020; McKenney et al. 2018; Sommer and Bäckhed 2013; Tuddenham and Sears 2015).

Within the microbiome there is thought to be a “core microbiome”, which is the set of microbial taxa that occurs consistently across multiple samples of a habitat or host (Hamady and Knight 2009). The taxa of the core microbiome are therefore expected to play a fundamental role in the ecosystem function of their habitat (Shade and Handelsman 2012; Turnbaugh et al. 2007). The core microbiome can be contrasted with the "transient microbiome", which is the array of taxa that changes over time based on factors including environmental parameters, diet, health and diurnal rhythms (Berg et al. 2020). Taxa assigned to the core microbiome typically make up only between 1% and 10% of the total microbiota diversity but constitute a high percentage of the microbial biomass (Custer et al. 2023).

Recent research across a variety of taxa have examined whether the genetics of a host, geography or ecological factors such as diet and habitat determine microbial community composition (Baldo et al. 2017; Grieneisen et al. 2019; Sevellec et al. 2019; Yeoh et al. 2017). Some studies have found significant association between microbial community relationships and the phylogeny of their hosts, a phenomenon defined as "phylosymbiosis" (Lim and Bordenstein 2020). Patterns of phylosymbiosis are sometimes attributed to vertical transmission, as microbiota may be transmitted across generations in both animals and plants (Eugene and Ilana 2016). Nevertheless, maternal transmission is thought to be far more limited in fish compared to other vertebrates (Funkhouser and Bordenstein 2013; Llewellyn et al. 2014). Phylosymbiosis may result from the different ecological niches inhabited by diversifying species; as observed in insects, mammals, plants and fish in both lab-controlled studies and in the wild (Amato et al. 2019; Bouffaud et al. 2014; Chiarello et al. 2018; Leigh et al. 2018; Ochman et al. 2010). Many others, however, have not found significant evidence for phylosymbiosis, instead noting that ecological factors better predict microbial community composition (Baldo et al. 2017; Bletz et al. 2017; Eichmiller et al. 2016; Gallo et al. 2020; Minich et al. 2021; Sylvain and Derome 2017; van Leeuwen et al. 2023). In aquatic animals more specifically, there is ongoing debate regarding the relative roles of the environment and genetics in shaping the microbiome, particularly as the aquatic environment can be seen to harbour its own microbiome due to the high density of inhabitants (Sehnal et al. 2021). Studies have shown that many gut microbial symbionts are horizontally transmitted from environmental water, which are particularly important for the initial microbial colonisation of the gut (Llewellyn et al. 2014; Sehnal et al. 2021; Sylvain and Derome 2017).

Among the ecological factors, many studies have pointed to diet being a major predictor of the microbiome, particularly in captive animals where feeding is easily controlled (Baldo et al. 2017; David et al. 2014; Martínez-Mota et al. 2020; Wilson et al. 2020). In mammals, the gut nutritional environment plays a major role in structuring the gut microbiota (Kohl and Dearing 2012; Ley et al. 2008; Schnorr et al. 2014), and a similarly important role has been suggested in fish (Baldo et al. 2017; Parris et al. 2019; Sevellec et al. 2019). Also, water conditions such as salinity, temperature and dissolved oxygen strongly influence fish microbial communities (Giatsis et al. 2015; Minich et al. 2021).

Compared to more widely studied taxa such as insects and mammals, research on fish gut microbial communities is in its infancy and biased towards aquaculture or economically important species such as carp, sturgeon, salmon and cod (Abdul Razak et al. 2022; Egerton et al. 2018; Keating et al. 2021; Zhu et al. 2021). Additionally, most studies have been concentrated on either a single species across multiple localities or multiple species in a single location, making it difficult to discern the relative roles of geography and genetics (Legrand et al. 2020; Sehnal et al. 2021).

1.1 | Study System

In this study we used fish of the genus *Cyprinella* (Leuciscidae) to explore relationships of the gut microbiome with genetic and environmental parameters. Commonly referred to as shiners, these fish are found in rivers, streams and lakes throughout North America and are generally small, with most species growing to less than 15 cm in length. *Cyprinella* typically have elongated, slender bodies and are often characterised by their silvery coloration and shiny, diamond-shaped scales (Page and Burr 2011). The genus encompasses 32 species, some of them with limited distributions, like the endangered Blue Shiner (*Cyprinella caerulea*) that only occurs in a limited drainage system; as well as widely distributed and abundant species such as the Blacktail Shiner (*Cyprinella venusta*) that occurs across much of the southeastern U.S. (Broughton and Gold 2000; Schönhuth and Mayden 2010). We focused on four species: the Red Shiner, *Cyprinella lutrensis* (Baird and Girard 1853); Blacktail Shiner, *Cyprinella venusta* (Girard 1856); Tricolour Shiner, *Cyprinella trichroistia* (Jordan and Brayton 1878); and Alabama Shiner, *Cyprinella callistia* (Jordan and Brayton 1877). Phylogenetically, these species represent a broad spectrum of the genus, with *C. lutrensis* and *C. venusta* being most closely related to each other, and *C. callistia* being sister to the rest of the species in the genus (Schönhuth and Mayden 2010). *Cyprinella venusta*, *C. trichroistia* and *C. callistia* are all native to the main study region in the Southeastern U.S. (Figure 1), with *C. venusta* inhabiting a very broad range and *C. trichroistia* and *C. callistia* more restricted, but still abundant in their native ranges (Schönhuth and Mayden 2010; Scott and Mayden 2008; Taylor and Gotelli 1994). Within the native range of these minnows, there is also one congeneric introduced species, *C. lutrensis*, that is one of the most broadly distributed and ubiquitous minnows of central North America (Matthews 1985). Widely introduced beyond its native range due to bait releases, *C. lutrensis* is known for its fast establishment rate that often allows it to outcompete native species (Schade and Bonar 2005). *Cyprinella lutrensis* is an ecological generalist, occupying a broad spectrum of stream types, and thriving in extremely harsh habitats (Matthews and Hill 1980). This is due in part to its ability to reproduce in a wide variety of microhabitats, and its high reproductive potential. The species matures rapidly and has the fastest intergenerational interval known for any North American minnow (Herrington and DeVries 2008; Marsh-Matthews et al. 2002; Vives 1993). Furthermore, *C. lutrensis* is known to hybridise with over 40% of its native regional congeners (Glotzbecker et al. 2016). One of the most well-known hybridisations is with the Blacktail Shiner, producing *C. lutrensis* × *C. venusta* hybrids that are fertile and capable of backcrossing with both parental species to produce F2 hybrids (Blum et al. 2010; Broughton et al. 2011; Hubbs and Strawn 1956). Due to these factors, *C. lutrensis* has been called the second most significant threat to the welfare of indigenous southwestern fishes (Dill and Cordone 1997).

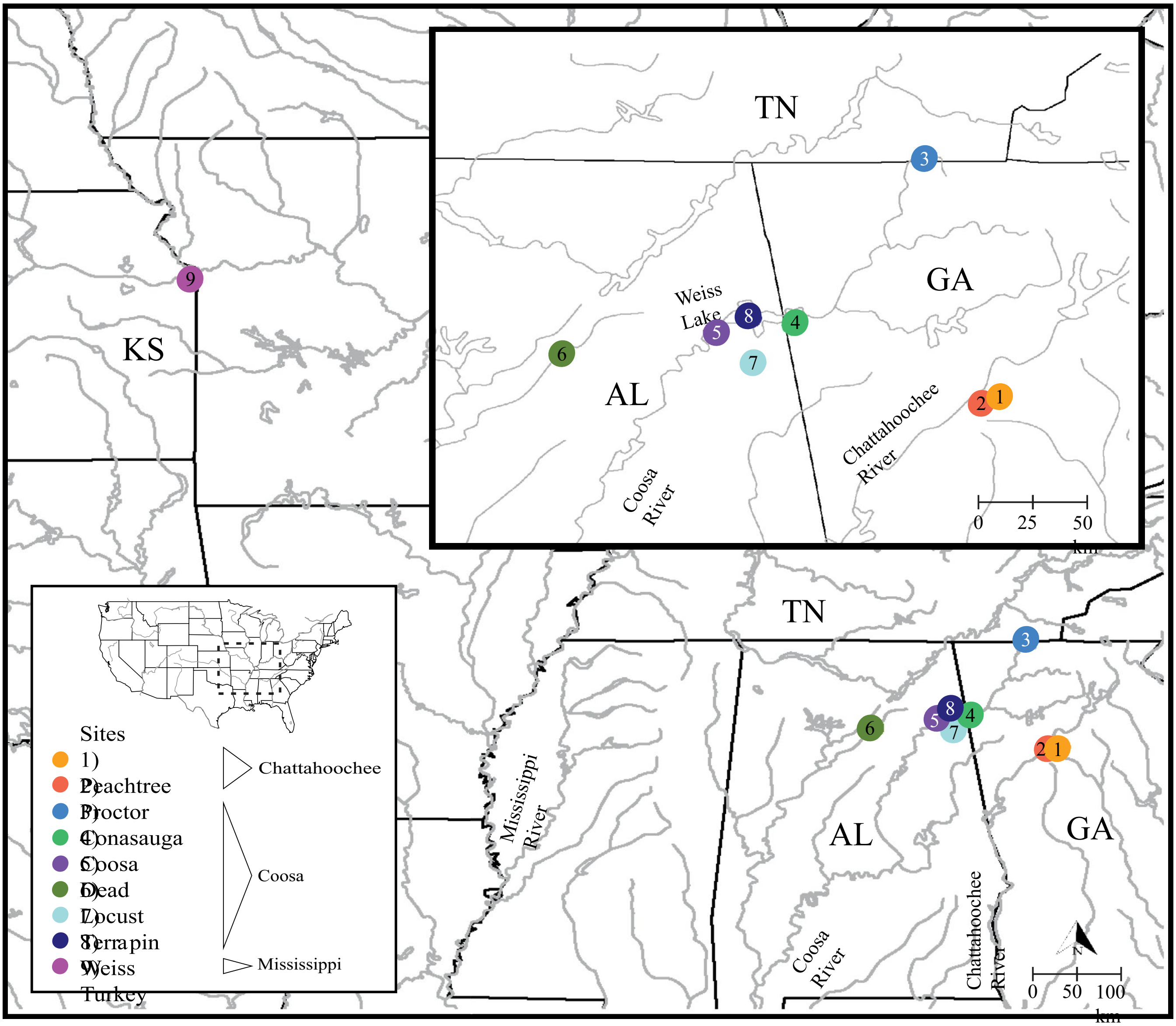

FIGURE 1 | Map of the U.S. showing sampling sites. Black dashed lines in the inset general map indicate the study area. Black solid lines indicate state borders, abbreviations correspond to states where samples were acquired. AL = Alabama, GA = Georgia, KS = Kansas, TN = Tennessee. Grey lines indicate rivers; the three watersheds sampled were the Mississippi (one site), Coosa (six sites) and Chattahoochee (two sites) basins. [Colour figure can be viewed at wileyonlinelibrary.com]

The purpose of this study is to investigate the relationships between the gut microbiota in closely related native and invasive fish species and their environment to understand the degree to which the microbiome exhibits environmental plasticity or is constrained by the genetics of the host. Specifically, we test two competing hypotheses: (1) gut microbial community composition is primarily explained by host species identity, thereby supporting the concept of phylosymbiosis; or (2) microbial communities are better explained by capture site or other environmental parameters, thereby suggesting plasticity or local adaptation. To address these hypotheses, we examined differences in bacterial phylotype richness and community composition in the gut microbiome of the regionally introduced *C. lutrensis*, and the native *C. venusta*, *C. trichroistia* and *C. callistia*, as well as the environmental aquatic microbiota across four U.S. states and three watersheds.

2 | Materials and Methods

2.1 | Sample Collection

We conducted field work across nine sites in Georgia, Tennessee, Alabama, and Kansas (U.S.) that encompassed the Coosa, Chattahoochee and Mississippi watersheds (Figure 1). Sites were selected based on known *C. lutrensis* introduction and hybridisation events, as well as public capture data (*FishMap.Org* 2015; Glotzbecker et al. 2016). Most sites were surveyed at least twice and consisted of a wide range of hydrological conditions, including small urban creeks, large reservoirs and rivers. We used a 2.5 m seine net to capture 93 individuals of target *Cyprinella* species and obtained usable data from 84 of them (see Results): 39 *C. lutrensis*, 21 *C. venusta*, 2 *C. lutrensis* × *C. venusta* hybrids, 9 *C. trichroistia* and 13 *C. callistia* (Table 1). We collected samples from sites where species co-occur, including *C. lutrensis* × *C. venusta* hybridisation zones, as well as where only *C. venusta* or *C. lutrensis* are present. We euthanised fish in the field with an overdose of MS-222, preserved them in ice, and immediately transported them to the laboratory for dissection of the gastrointestinal tract (< 4 h). We separated the stomachs from the intestine and stored them separately at −20°C. We also collected pectoral fin clips from each fish and preserved them in ethanol at −20°C for DNA analysis. We measured the total body length (TBL) of each specimen and noted any males with breeding coloration and gravid females. All samples were collected and processed in accordance with Federal and State regulations.

We collected 1 L of water from each site at the same time as the fish sampling for the analysis of environmental microbiota. Seven water samples were immediately filtered in the field using a Millipore Stericup Quick Release Vacuum Driven Disposable Filtration System with a 0.22 μm filter membrane and a manual vacuum pump and the filter paper was preserved frozen at −20°C until extraction. The remaining two samples of water were first frozen and then thawed and filtered in the laboratory using the same device.

2.2 | DNA Extraction and Amplification

To confirm the morphological species identifications, we extracted total DNA from the host fin clips using the Qiagen DNeasy Blood & Tissue Kit (Qiagen Inc., Valencia, CA) and used 1 μL of the eluted DNA to amplify the mitochondrial cytochrome *b* in 25 μL PCR reactions containing: 1X PCR buffer, 2 mM $MgCl_2$, 2 mM each dNTP, 0.625 U of *Taq* polymerase (New England BioLabs, Ipswich, MA), and 0.5 μM of each primer. We used primers GLU and THR (Schmidt et al. 1998) and the following thermocycling conditions: 4 min at 94°C, 35 cycles of 45 s at 94°C, 1 min at 52°C, and 1 min 15 s at 72°C and a final step of 5 min at 72°C. We purified PCR amplicons using Exonuclease *I* and Shrimp Alkaline Phosphatase (New England BioLabs, Ipswich, MA) enzymatic reactions and sequenced them using both amplification primers at Macrogen-Psomagen sequencing facilities. The chromatograms were trimmed and assembled using Geneious v.2022.2.2 and identified by blasting to the NCBI nr database. All sequences are available under GenBank accession numbers PQ030722-PQ03080.

To sequence the bacterial contents in the gut, first we physically and chemically lysed the whole intestine samples using the microbead tubes in the Qiagen DNeasy PowerSoil DNA Isolation Kit (Qiagen Inc., Valencia, CA). Similarly, to sequence the contents in the water samples, we cut the filter paper into small slivers of approximately 5 × 5 mm before lysing the organic material on the paper. Second, we conducted DNA extractions using the same kit according to the manufacturer's protocol. We then amplified the 16S ribosomal RNA (rRNA) gene commonly used in bacterial metabarcoding studies (Johnson et al. 2019). We amplified the V3 region and most of the V4 region of the 16S rRNA gene, nucleotides 341–785, using primers 341b (5′-CCTACGGGNGGCWGCAG-3′) and 785a (5′-GACTACHVGGGTATCTAATCC-3′) (Herlemann et al. 2011) attached to Illumina PE adaptors. The 10 μL PCR reactions consisted of 5 μL Platinum Hot Start PCR Master Mix (Thermo Fisher Scientific, Waltham, MA), 2 μL enhancer, 0.25 μL of each primer (10 μM) and 2.5 μLnuclease-free H_20. Thermocycling conditions were as follows: 5 min at 95°C, 25 cycles of 40 s at 95°C, 2 min at 55°C, 1 min at 72°C and a final step of 7 min at 72°C. A negative control was included for all PCR reactions.

Following this initial PCR step, we diluted the amplicons 1:10 before the barcoding PCR and quantified their concentration using Qubit (Thermo Fisher Scientific, Waltham, MA). In this step, we constructed libraries by attaching a unique 8-bp forward/reverse barcode pairing to each of the samples using i7 and i5 TruSeq adapters from IDT (Integrated DNA Technologies Inc.) (Miya 2022). The reaction volume of 25 μL consisted of 12.5 μL Platinum Hot Start PCR Master Mix, 0.75 μL of each index (15 μM), 5 μL of template amplicon, and 6 μL

TABLE 1 | Specimens captured from each species at each site. Site name and number correspond to Figure 1. In total, 39 *C. lutrensis*, 21 *C. venusta*, 2 *C. lutrensis* × *C. venusta* hybrids (*C. lut* × *ven*), 13 *C. callistia* and 9 *C. trichroistia* were captured across 9 sites.

Site	Watershed	Coordinates (Lat/Long)	*C. lutrensis*	*C. venusta*	*C. lut* × *ven*	*C. callistia*	*C. trichroistia*
(1) Peachtree	Chattahoochee	(33.81, −84.36)	9				
(2) Proctor	Chattahoochee	(33.78, −84.44)	1				
(3) Conasauga	Coosa	(35.01, −84.73)		4		3	3
(4) Coosa	Coosa	(34.17, −85.40)		4			
(5) Dead	Coosa	(34.13, −85.79)	1	4	1		
(6) Locust	Coosa	(34.02, −86.57)		3		5	
(7) Terrapin	Coosa	(33.98, −85.60)				5	6
(8) Weiss	Coosa	(34.20, −85.62)	15	6	1		
(9) Turkey	Mississippi	(39.03, −94.69)	13				
Total			39	21	2	13	9

H_20 (given amplicon concentration of 1 ng/μL). Thermocycler conditions were as follows: 3 min at 95°C, 15 cycles of 20 s at 98°C, 15 s at 72°C and a final step of 5 min at 72°C. We used 0.8× of a SPRI beads substitute (Rohland and Reich 2012) to clean up the libraries, quantified them in Qubit, and combined them at equimolar concentrations in pools of 10 that were sent off for sequencing by Novogene, U.S.A in one lane of Illumina MiSeq PE250.

2.3 | Computational Analysis

After receiving the demultiplexed sequences with primers and adapters removed, we imported the raw reads into the QIIME2 platform version 2022.11 (Bolyen et al. 2019) for sequence analyses and processing. QIIME2 was accessed through the UTC SimCenter Epyc Cluster. Further sequence quality control and generation of amplicon sequence variants (ASVs) were completed using the DADA2 V1.4 pipeline (Callahan et al. 2016) with default settings. For the 16S rRNA gene data, we only kept the forward reads due to insufficient quality of the reverse reads, and we trimmed the reads between position 100 and 247 base pairs (bp) to obtain a total of 147 bp. The taxonomy of each ASV was determined with BLAST against the SILVA database number 99, version 138.1 (Quast et al. 2013). After carefully checking the taxonomic assignment for each ASV, we discarded those that were taxonomically assigned to mitochondria, chloroplasts, as well as the unassigned ASVs. We also removed all ASVs that were present in only two or fewer samples as well as the ones represented by less than 10 reads. Three ASVs that were present in a negative control sample were also removed from the dataset.

2.4 | Variables Analysed

In our analysis of the diversity and structure of the gut microbiome, we examined three key response variables. The first two, species richness and phylogenetic diversity (PD), are alpha diversity metrics that represent the biodiversity within a sample. The third response variable was beta diversity, as represented by community composition, or the distribution of ASVs among explanatory variables. To test the factors shaping the diversity and structure of the gut microbiome, we primarily examined the species of host fish and sampling as key explanatory variables. We additionally examined an array of secondary explanatory variables such as sampling session, sex of fish and total body length (TBL).

2.5 | Core Microbiome

Differentiating between the core and transient components of the microbiome can be useful in determining the levels of fidelity between microbe and host. To determine the core microbiome of each of the four species of fish studied, as well as for the total data set, we used the following methodology recommended in Custer et al. (2023). In Microsoft Excel, we retained all available abundance data for each ASV and grouped the dataset first by species, then by site. An ASV was considered to belong to the core microbiome if it was found in at least one sample from each site in a given species group and present in at least 50% of the total fish of that group.

2.6 | Environmental (Aquatic) Microbiome

For the environmental microbiota, we sorted ASVs based on occurrence per site. We then used a network analysis to visualise correlation of microbiota ASVs between water samples and fish gut microbiome samples. To do this, we generated a Spearman's correlation matrix applying a correlation threshold of ≥ 0.5, and a p-value cutoff of > 0.05, using function 'rcorr' in the R package 'hmisc' (R Core Team 2023; Harrell Jr and Harrell Jr 2019). The resulting network was plotted using CYTOSCAPE v3.10.0, with nodes colour-coded by sampling site (Shannon et al. 2003). We also tested for correlation between water sample prokaryotic beta diversity (Jaccard dissimilarity index) and geographic distance between sites (km) via a Mantel test (function 'mantel.rtest' in the package 'ade4') (Dray and Dufour 2007).

2.7 | Alpha Diversity

We calculated mean microbial alpha diversity to understand what explanatory variable best explained differences in microbial diversity among groups. Microbial alpha diversity was assessed as Faith's phylogenetic diversity (PD) values (Faith 1992) and ASV richness. Phylogenetic diversity was calculated using the function 'pd' in the R package 'picante' 1.8.2 (Kembel et al. 2010). We calculated species richness for each 16S rRNA gene sample in Microsoft Excel, as defined by the count of ASVs present in a given sample. We then fit linear models with PD and ASV count as the response variables and tested the significance of explanatory variables using the function 'lm' in the R package 'stats' 3.6.2. We ranked the models for best fit using the function 'aictab' in the package AICcmodavg (Mazerolle 2023). To test whether there was a significant difference in 16S rRNA gene alpha diversity among different sites and species, we carried out an analysis of variance (ANOVA) using the function 'anova' in the package 'stats'.

2.8 | Beta Diversity

To test whether there was a significant difference in 16S rRNA gene community composition among different sites and species, we performed a permutational multivariate analysis of variance (PERMANOVA) using the 'adonis2' function in the R package 'vegan' (Oksanen 2010). Community composition distance matrices were calculated using the Jaccard presence/absence dissimilarity index. Presence/absence-based metrics are generally recommended for microbial community analyses rather than abundance-based metrics to mitigate biases introduced by variations in 16S rRNA gene copy number across taxa, particularly when absolute abundances are unknown (Větrovský and Baldrian 2013). Therefore, we consistently used Jaccard distances throughout our analyses to ensure robust comparisons.

To evaluate potential contamination or batch effects, we tested for the impact of extraction and amplification batches

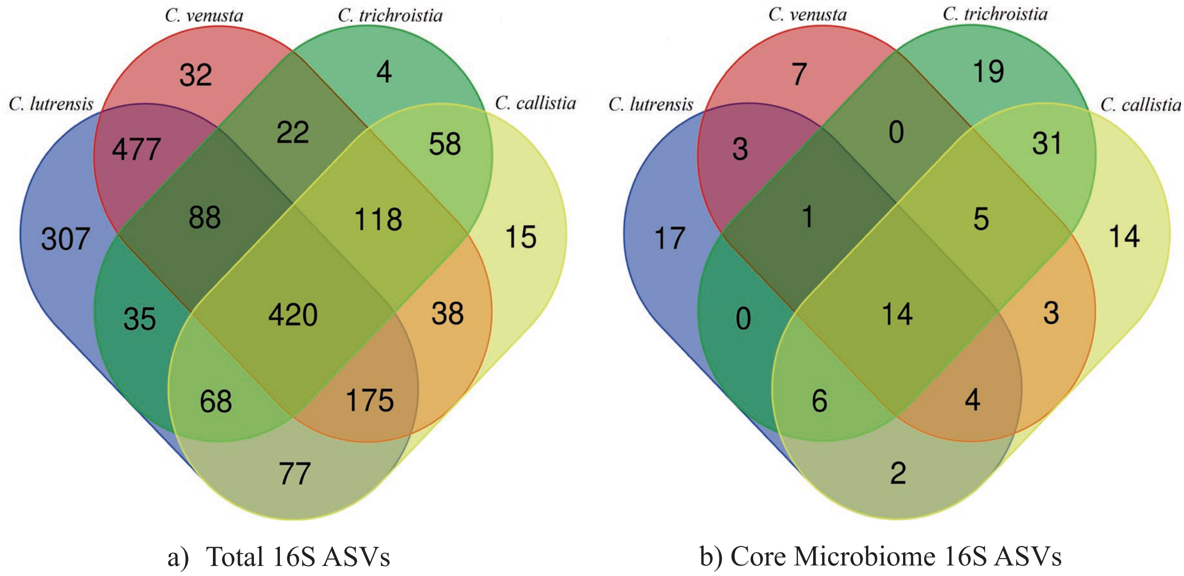

FIGURE 2 | Venn diagrams showing (a) total number of 16S ASVs recovered from each species of *Cyprinella*, and (b) number of 16S ASVs per species of *Cyprinella* fitting the criteria of core microbiome. [Colour figure can be viewed at wileyonlinelibrary.com]

as independent variables via PERMANOVA. We also used the 'betadisper' function in 'vegan' to assess differences in beta diversity dispersion among host groups.

We conducted a non-metric multidimensional scaling (NMDS) analysis to visualise dissimilarities among microbial communities. We performed these tests for the complete data set of gut microbiomes of all *Cyprinella* species, and for specific subsets of the data: (1) *C. lutrensis* populations, (2) Weiss Lake populations and (3) native Southeastern *Cyprinella* species (i.e., *C. callistia, C. trichroistia, C. venusta*). The goal of this approach was to tease apart nested variables and understanding the relationship between community composition and invasiveness.

To test for phylosymbiosis, we performed a Mantel test to assess the correlation between host genetic distance (Tamura-Nei, based on mitochondrial cytochrome *b*) and prokaryotic beta diversity (Jaccard dissimilarity index) (Lim and Bordenstein 2020; Tamura and Nei 1993). Finally, we used a network analysis to visualise differences in gut microbiota among individual fish specimens. The resulting network was plotted using CYTOSCAPE v3.10.0, with nodes colour-coded by sampling site and labelled by species.

3 | Results

3.1 | Presence of Native and Invasive Species and Their Hybrids

We found native *C. venusta* at five sites across two watersheds, co-occurring with at least one other species of *Cyprinella* in all but one of these sites. The Conasauga river exhibited the greatest diversity of native *Cyprinella*, including *C. venusta*, *C. trichroistia*, and *C. callistia*. *Cyprinella trichroistia* and *C. callistia* were also found without the presence of *C. venusta*, but always co-occurred with at least one other *Cyprinella* species (Table 1).

We found invasive *C. lutrensis* at four sites across two watersheds. Two of these sites were at Weiss Lake, Alabama: one site (Weiss) is a lacustrine environment above the dam, and the other is a slightly more riverine section below the dam known as Dead River. At each site we also found *C. venusta*, and we captured four fish that morphologically resembled *C. venusta* with a distinct black spot on the tail, but had a *C. lutrensis* mitochondrial haplotype, indicating possible hybridisation. We did not find *C. lutrensis* to co-occur with any *Cyprinella* species other than *C. venusta*. At the two sites in the Atlanta metropolitan area, we found *C. lutrensis* without any other *Cyprinella* species, and as the most abundant fish in the assemblage. Within its native range, we found *C. lutrensis* to be the only representative of its genus at the Turkey Creek site in the Mississippi watershed (Table 1).

3.2 | Metabarcoding of Gut Microbiome

We successfully recovered 16S rRNA gene reads for 84 of the 93 specimens captured. For eight samples, we were unable to amplify the 16S rRNA gene, and for one sample no reads were recovered. We initially sequenced a total of 5,902,276 reads that clustered to 11,264 ASVs. After quality control checking, the final dataset consisted of a total of 2,714,395 reads and 2039 ASVs, with a mean of 30,159 reads per sample (minimum: 2193; maximum: 146,138 reads), and of these 2039 ASVs, 420 were found in all four fish species (Figure 2).

3.3 | Metabarcoding of Aquatic Microbiome

The microbial composition of the 9 water samples was significantly different from that of the fish samples (PERMANOVA, $R^2 = 0.057$, $F = 5.397$, $p < 0.001$). Out of the 2039 total ASVs found across all samples, 1573 were present only in fish, 103 ASVs were found only in water, and 363 ASVs were found in both fish and water samples (Figure S1). Only 10% of the ASVs recovered from

TABLE 2 | Sixteen most common gut microbiota classes found in all fish samples analysed. Firmicutes, Actinobacteria and Proteobacteria were most dominant phyla detected.

Phylum	Class	Abundance	% total reads
Firmicutes	Clostridia	851,110	31%
Actinobacteria	Actinobacteria	421,909	16%
Firmicutes	Bacilli	379,486	14%
Proteobacteria	Alphaproteobacteria	374,027	14%
Proteobacteria	Gammaproteobacteria	225,203	8%
Planctomycetes	Planctomycetacia	133,093	5%
Cyanobacteria	Oxyphotobacteria	105,105	4%
Actinobacteria	Acidimicrobiia	100,555	4%
Actinobacteria	Thermoleophilia	46,676	2%
Chloroflexi	KD4-96	26,248	1%
Verrucomicrobia	Verrucomicrobiae	21,184	1%
Chloroflexi	Anaerolineae	17,643	1%
Actinobacteria	Coriobacteriia	13,345	< 1%
Chloroflexi	Chloroflexia	5223	< 1%
Proteobacteria	Deltaproteobacteria	4150	< 1%
Patescibacteria	Saccharimonadia	3790	< 1%

the water samples were found across the majority of samples, indicating a high degree of microbial variability among water bodies. Two ASVs were universally present across all water samples, both belonging to uncultured bacterial families associated with plant roots. These two ASVs were also found in the majority of fish samples, suggesting their widespread presence in the environment.

Based on the network visualisation, there was no clear relationship found between the prokaryotic community composition of a water sample and the community composition of fish gut microbiomes found at that site (Figure S2). Additionally, the community composition of water samples was not significantly correlated with the geographic distance between sites (Mantel test 9999 replicates, $p = 0.305$). The two samples that were frozen before filtering had substantially lower ASV counts (mean = 80.3) than the other seven samples (mean = 140.2). We repeated the analyses without these two samples and obtained similar results.

3.4 | Core Microbiome

The abundance of gut microbiota (i.e., number of reads) across all fish species sampled was dominated by 16 prokaryotic classes (Table 2). These 16 classes were present in at least 80% of the samples, suggesting that they broadly contain the "core microbiome" of these species. At the species level, 37 ASVs met the criteria of core microbiota for *C. venusta*, 47 for *C. lutrensis*, 78 for *C. trichroistia* and 81 ASVs for *C. callistia*. Fourteen ASVs were shared between the majority of fish in all species, making these candidates to constitute a "core microbiome" for the genus *Cyprinella* (Table S1). Notably, the overlap between the species core ASVs was proportionally much lower than between the total of ASVs recovered, suggesting a greater uniqueness of the core microbiome communities (Figure 2). Note that *C. lutrensis* and *C. venusta* were captured from a broader range of sites, therefore making the threshold for considering an ASV to be "core" more stringent.

3.5 | Prokaryotic Alpha Diversity

The trends in values of alpha diversity measured as Faith's PD values and ASV richness broadly mirrored each other. Overall, species showed similar diversity values, with *C. venusta* showing the largest within-species variation, despite being a smaller sample size than *C. lutrensis*. Across sites, the four *C. venusta* samples from the Coosa River showed the lowest diversity. Meanwhile, all the samples from Conasauga and Weiss (which included *C. venusta*) showed some of the highest diversity (Figure 3a,b). When examining only the prokaryotic diversity of *C. lutrensis*, there is an observable trend where the alpha diversity values of the fish microbiota in the native range were higher than in fish from the invasive range (Figure 3c,d). Native *C. lutrensis* from the Mississippi watershed had significantly higher ASV richness than invasive fish from the Coosa and Chattahoochee watersheds (two sample *t*-test, $p = 0.015$). However, phylogenetic diversity was not significantly different between the invasive and native groups ($p = 0.069$).

The linear models with the best fit (based on AIC scores) for explaining both phylogenetic diversity and ASV richness were the ones consisting of both site and sampling session, and with

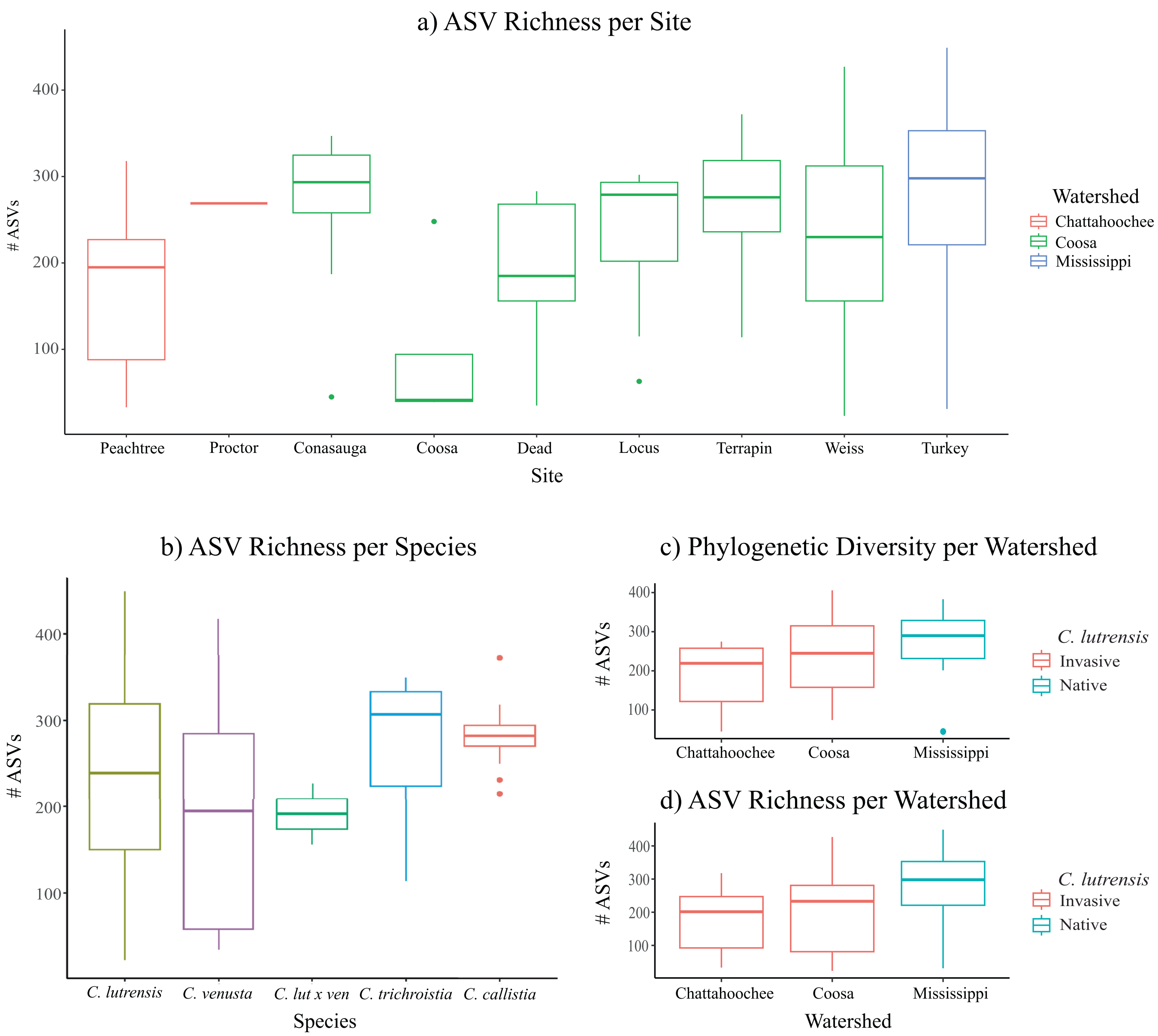

FIGURE 3 | Boxplots of alpha diversity metrics of ASV richness across sites (a) and species (b), and alpha diversity metrics of phylogenetic diversity (c) and ASV richness (d) across native and invasive *C. lutrensis* populations. Native *C. lutrensis* from the Mississippi watershed had significantly higher species richness than invasive fish from the Coosa and Chattahoochee watersheds (two sample t-test, $p = 0.015$). However, phylogenetic diversity was not significantly different between the invasive and native groups ($p = 0.069$). [Colour figure can be viewed at wileyonlinelibrary.com]

sampling session as the sole explaining variable. When we did not include sampling session, site became the strongest explanatory variable of gut microbial diversity (Site AIC weight = 0.81, Species AIC Weight = 0.19). The ANOVA tests indicated that site and sampling session explain a significant proportion of the variance of phylogenetic diversity of the gut microbiome and its ASV richness, while host species and TBL do not. Additionally, the sex of the fish explained a significant proportion of ASV richness, but not of phylogenetic diversity (Table 3).

3.6 | Community Composition

Overall, we found that microbial community composition is significantly different among sites (PERMANOVA, $R^2 = 0.338$,

TABLE 3 | Statistical results for ANOVA tests of alpha diversity metrics Richness and Phylogenetic Diversity.

	Richness		Phylogenetic Diversity	
Explainer	***F*-value**	***p***	***F*-value**	***p***
Site	2.514	0.0178*	2.446	0.0209*
Species	1.739	0.1499	0.972	0.4276
Session	3.16	0.0012*	3.309	0.0007*
Sex	4.932	0.0307*	3.227	0.0782
TBL	0.032	0.8588	0.703	0.4042

Note: *P*-values below the significance threshold of 0.05 are indicated with *.

$F = 4.716$, $p < 0.001$), and among species ($R^2 = 0.153$, $F = 3.529$, $p < 0.001$). Despite site explaining over two times the variance than host species (33% and 15%, respectively), species was both significant when looked at independently and when nested within the site variable, suggesting that microbial communities differ among species found at the same site. The interaction of site and species was non-significant, implying that the two variables are additive rather than interactive. Analysis of multivariate homogeneity of group dispersions revealed significant differences in dispersion across sites (betadisper, $F = 5.326$, $p < 0.001$) and host species ($F = 14.805$, $p < 0.001$), indicating that some host groups exhibit greater variability in microbial community composition. In contrast, no significant differences were detected among randomly grouped samples extracted and amplified in batches (PERMANOVA, $R^2 = 0.09248$, $F = 0.9426$, $p = 0.684$).

The influence of host species was further supported by the Mantel test, which revealed a significant positive correlation between host genetic distance and microbial community composition (Mantel test, $r = 0.084$, $p = 0.038$; Figure 4). This finding suggests an evolutionary control in shaping gut symbiont communities.

The strongest explainer of community composition was sampling session ($R^2 = 0.426$, $F = 4.330$, $p < 0.001$). For example, the Weiss Lake site was sampled four separate times, and the microbial community composition was significantly different between each of those sampling days ($R^2 = 0.294$, $F = 3.960$, $p < 0.001$). Total body length also explained a small but significant portion of the community composition variability ($R^2 = 0.026$, $F = 2.180$, $p = 0.030$). There were not significant differences in the community composition based on the sex of fish ($p = 0.124$).

The NMDS plots confirmed that differences in community composition follow notable grouping along lines of site, watershed and sampling session (Figure 5). The network analysis also supported the statistical tests and showed a pronounced differentiation in intestinal microbiota composition between capture sites (Figure 6). Fish gut microbiomes tended to be most similar to those found at the same site, and secondarily similar to those of the same species. More broadly, there appears to be clustering along the boundaries of watershed as well. For example, the fish microbiota from the Conasauga river and Terrapin creek sites, both of which belong to the Coosa watershed, are notably similar despite the large geographic distance between the sites.

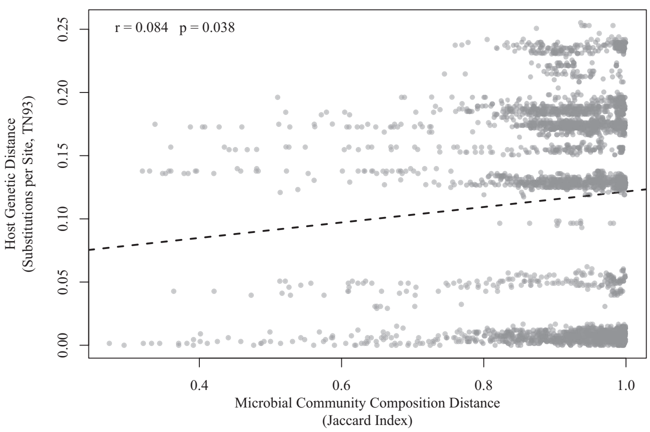

FIGURE 4 | Relationship between host genetic distance and microbial community composition. The scatterplot shows the significant positive correlation between host genetic distance (measured as substitutions per site using the TN93 model) and microbial community composition distance (Jaccard index). Each point represents a pairwise comparison between hosts. The dashed black line represents the best-fit linear regression model.

4 | Discussion

In this study, we found that host species and geographic location have a combined effect in shaping the gut microbial community composition of minnows in the southeastern U.S., but only sampling site has an effect on the alpha diversity of their microbiota. Sampling site, which includes both distinct geographic and ecological conditions, was consistently a more powerful explainer of microbial composition, suggesting that at the observed level of phenotypic and genetic distance (i.e., intra-genus), fish microbial composition is highly plastic and primarily driven by the environment. We also incorporated a regionally invasive species and found that the microbial alpha diversity was higher in native populations than invasive, hinting to a possible impact of the founder effect stemming from colonisation on microbial diversity.

4.1 | Phylosymbiosis or Something Else?

Studies in vertebrates posit that patterns of phylosymbiosis are more pronounced at broad evolutionary scales (Groussin et al. 2017). However, as we zoom in on the tree of life, a grey area emerges where species and populations share similar niches and life histories, making phylosymbiosis patterns difficult to discern (Brooks et al. 2016; van Leeuwen et al. 2023). Identifying where these boundaries lie can improve our understanding of how the microbiome is derived and controlled, as well as the relative roles of horizontal and vertical transmission of microbes in driving phylosymbiosis (Lim and Bordenstein 2020).

In this study, we found evidence of phylosymbiosis within the genus *Cyprinella*. While the collection site was the strongest predictor of microbial community composition, host species also had a significant effect. This pattern was formally tested via a Mantel test, which revealed a significant positive correlation between host genetic distance and microbiome composition. Based on these results, we propose that differences in gut microbiome composition among host species are primarily due to changes in community turnover, with fish gut microbiota assembled more by environmental filtering than dispersal limitation. This interpretation is further supported by the lack of significant geographical correlation in microbiome beta diversity. Interestingly, other studies have found that environmental filtering may change during host development, which is also consistent with the temporal changes that we observed (Yan et al. 2016).

Although our results support phylosymbiosis, we must acknowledge the limitations of this correlation and the data available. First, environmental or stochastic variables remain stronger explainers of microbial community composition. Second, the significance of species as an explanatory variable may reflect differences in habitat and diet, or drift-driven genetic differences that are not necessarily correlated with

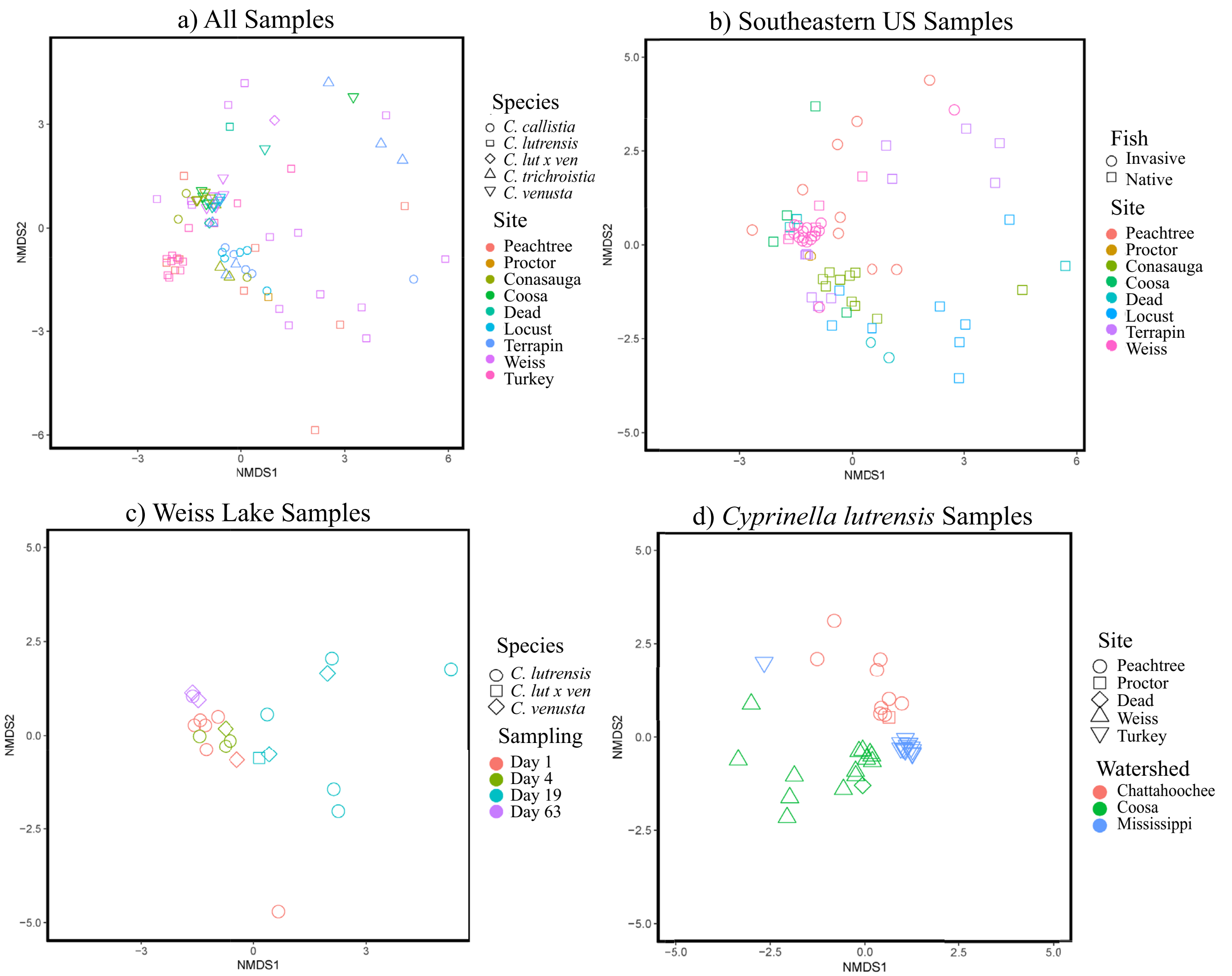

FIGURE 5 | NMDS ordination plots showing the effects of species and site on microbial community composition. From top left to bottom right: (a) all samples coloured by site and shaped by species; (b) samples from sites in the southeastern U.S. (not including Mississippi watershed); (c) samples from Weiss Lake, where *C. lutrensis* × *venusta* hybridisation occurs. (d) *C. lutrensis* samples only, Chattahoochee and Coosa specimens are invasive while Mississippi are native. [Colour figure can be viewed at wileyonlinelibrary.com]

evolutionary time. Therefore, these results may also fit a scenario of horizontal transmission controlled by ecological factors, as has been suggested in similar studies for other species like Zebrafish (Sylvain and Derome 2017), Whitefish (Sevellec et al. 2019), European Seabass (Kokou et al. 2019) or Kingfish (Minich et al. 2021).

4.2 | Geographical and Ecological Drivers

Given that we found only a small amount of the microbial community to be driven by vertical transmission, the next question is how else the microbiome might be derived. In our results, two explanatory variables were clearly the strongest: site and sampling session. These two variables are intertwined both with each other, as some sites were sampled only once, and with other variables, including water quality, food availability and habitat type, all of which influence fish gut microbiota (Baldo et al. 2019; Eichmiller et al. 2016; Gallo et al. 2020). Geographic distance did not significantly correlate with community composition, and therefore it can be assumed that ecological rather than geographical factors drive microbial recruitment. Sampling session being a stronger explainer than site could imply that temporal changes in life stage, diet and/or water quality drive a substantial portion of the community composition and suggest a greater effect of phenotypic plasticity on the structure of the gut microbiome than of local adaptation (Baldo et al. 2019; Minich et al. 2021; Parris et al. 2019; Sylvain and Derome 2017). Finally, the small, yet significant amount of community composition explained by the body length of the fish may also be related to the previous statement and point to life stage and diet being major drivers, as the diet of *Cyprinella* changes with age and size (Scott and Mayden 2008; Yan et al. 2016).

Preliminary data from 18S rRNA gene metabarcoding of fish gut content, a method recommended for characterising the diet of fish (Albaina et al. 2016), revealed a significant correlation between host diet and gut microbial composition. However, the conclusions drawn from these findings are limited by a low number of reads and the inherent link between dietary content

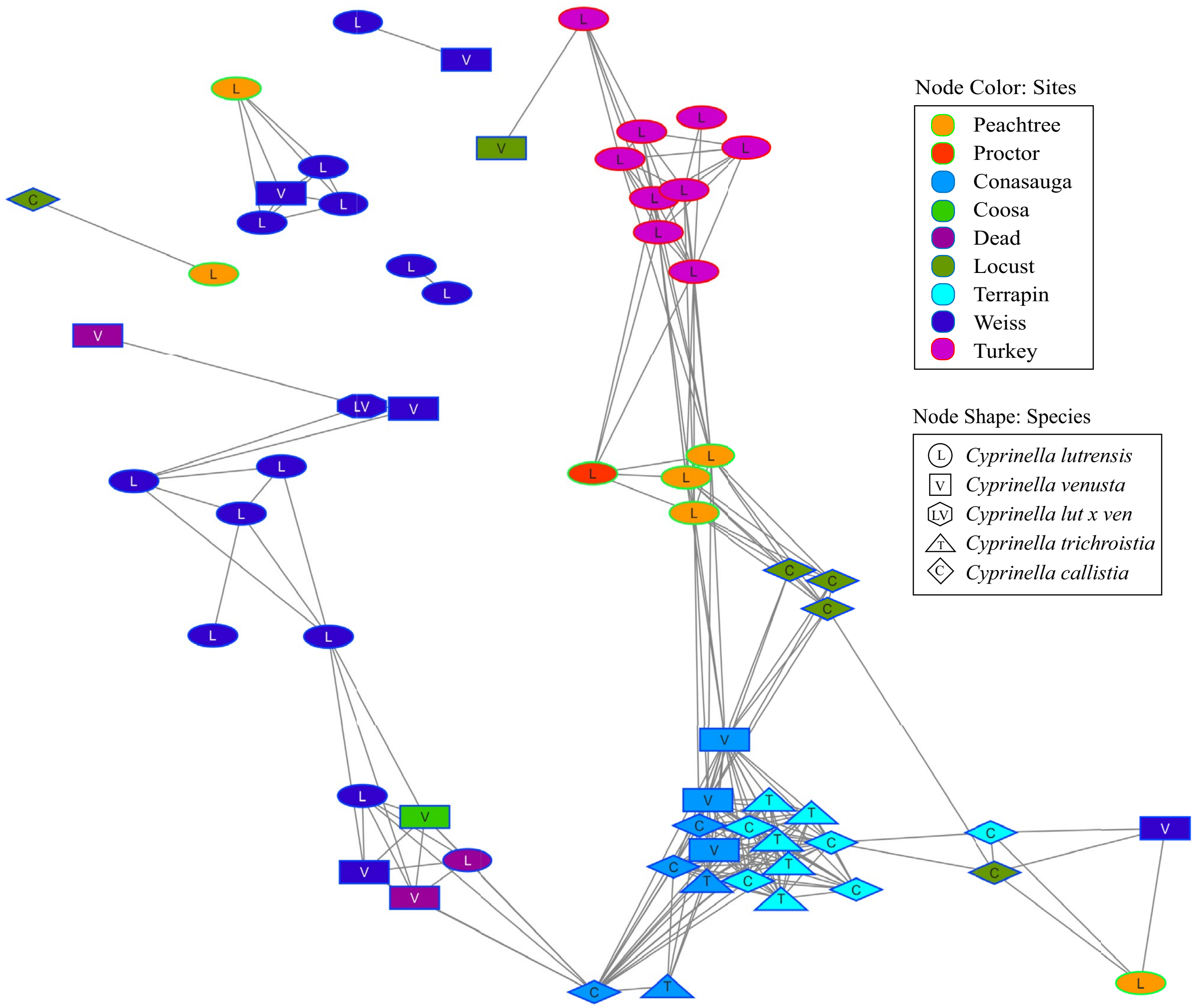

FIGURE 6 | Network analysis of intestinal microbiota of *C. lutrensis* (L), *C. venusta* (V), *C. callistia* (C) and *C. trichroistia* (T). Each node represents the collection of microbiota within a fish, with the colour and shape of the node representing the corresponding site of collection. The connecting lines represent significant Spearman index correlations, with the length of the line representing strength of the correlation (edge-weighted spring-embedded layout). [Colour figure can be viewed at wileyonlinelibrary.com]

and associated microbes, requiring more thorough investigation. A summary of the eukaryotic genera identified in the gut samples and their potential correlation with microbial ASVs is provided in Table S2 and Figure S3.

While we did not find a statistical correlation between the water samples and the fish microbiota recovered from those water bodies, we noted that a majority of microbial ASVs recovered from the water samples were also present within at least one fish sample. This indicates that some microbiota may be accumulated from the environment (Figure S2), supporting the hypothesis that waterborne microbes play a key role in gut microbiome recruitment, with additional factors such as drift and dispersal further shaping microbial diversity (Burns et al. 2015; Llewellyn et al. 2014; Sylvain and Derome 2017).

Microbes may also be horizontally transferred among fish, as *Cyprinella* often form schools (Taylor and Gotelli 1994). However, we did not observe significant differences between the gut microbiota of *C. lutrensis* and *C. venusta* at Weiss, where they co-occur, school together and hybridise. Schooling behaviour and other social interactions may influence microbial composition, yet disentangling their effects on the microbiome remains challenging without controlled experimental conditions.

4.3 | Microbiome Composition and Core Microbiome

The microbial diversity of *Cyprinella* was most dominated by Firmicutes (45%), Actinobacteria (22%) and Proteobacteria (22%), which is similar to other freshwater fish studies (Sevellec et al. 2019; Zeng et al. 2020). Clostridia, the most common bacterial class, has been found to be relatively abundant in cyprinid fish (Kim et al. 2021). Our results indicate that the gut microbiome of *Cyprinella* individuals is highly dynamic and influenced by many variables, with no single factor consistently explaining more than 30% of the microbial community composition. Notably, "sampling session", which includes both spatial and temporal variation, was consistently a stronger explainer than site alone. This is further supported by the fact that only a small percentage of the microbial ASVs meet the threshold to be considered core microbiota.

For *C. lutrensis* and *C. venusta*, only ~2% of ASVs were shared among the majority of individuals across all sites, meaning that the remaining 98% of microbial taxa are largely transient. For *C. trichroistia* and *C. callistia*, this proportion was slightly higher (about 5%), which may be due to their more restricted geographic distribution, resulting in a less stringent threshold for defining core microbiota. These results are congruent with previous research on fish microbiomes, for example on European Seabass, that only found eight ASVs present in at least 90% of specimens (Kokou et al. 2019).

More broadly, our findings are consistent with the idea that taxonomically richer microbiomes imply greater stochasticity in community assembly (Mallott and Amato 2021). The transient nature of the gut microbiome in these fish raises further questions about the role of vertical transmission in aquatic organisms, as our study suggests that the aquatic environment may exert a stronger influence on the intestinal microbiota composition compared to terrestrial systems (Zeng et al. 2020).

4.4 | Invasion and the Microbiome

The gut microbiome is thought to play an essential role in host fitness and local adaptation to new environments, leading to the hypothesis that microbiome composition may also be a determinant of invasion success (Dulski et al. 2020; Lefort et al. 2017; Ley et al. 2008; Rennison et al. 2019; Zhu et al. 2021; Zilber-Rosenberg and Rosenberg 2008). At the same time, colonisation of a new environment may drive shifts in the microbiome of an invading population. Many studies have found that microbiome composition is most strongly influenced by environmental factors or diet, both of which can change significantly in a novel habitat (Baldo et al. 2017; Bletz et al. 2017; Eichmiller et al. 2016; Gallo et al. 2020; Minich et al. 2021; Sylvain and Derome 2017; van Leeuwen et al. 2023). Furthermore, the founder effect—where introductions result in a reduction of genetic variability and non-random sampling of alleles relative to the source population—may similarly affect the microbiome, potentially leading to lower microbial alpha diversity in the introduced population, at least temporarily (Dlugosch and Parker 2008; Millie and Susanna 2021). Consistent with this expectation, our results show a reduction of alpha diversity in invasive *C. lutrensis* populations, which exhibited lower microbial phylogenetic diversity and species richness compared with the native *C. lutrensis*. However, alpha diversity in *C. lutrensis* was not significantly different from that of the three native *Cyprinella* species, and therefore not sufficiently supporting the hypothesis that the diversity of the gut microbiome may directly contribute to invasion success. Additionally, within Weiss Lake samples, where invasive *C. lutrensis* co-occurs with native *C. venusta*, we found no significant differences in microbial community composition or alpha diversity between the two species. This is congruent with our broader finding that sampling site, rather than host species, was consistently the strongest predictor of microbial composition and diversity.

Broadly, our study highlights the utility of introduced populations as natural models for studying the founder effect and its potential impacts on the microbiome. Future explorations could expand upon this model both taxonomically and temporally to untie confounding variables and provide a holistic perspective on how microbiomes respond and influence species introductions.

Author Contributions

W.H.-R. and F.A. conceived the study. W.H.-R. collected samples. W.H.-R. and F.A. performed the lab work. W.H.-R. and F.L. performed computational and statistical analyses. W.H.-R. led the writing and the rest of the authors contributed to edit and review the manuscript.

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Disclosure

Benefits generated: This article reflects the work carried out by William Hanson-Regan to obtain his Master of Science degree at the University of Tennessee at Chattanooga. Also, three students from the University of Tennessee at Chattanooga participated in this project and acquired undergraduate research experience. All data, scripts, and results have been shared with the broader public in appropriate public repositories, as described above.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Raw sequences of 16S and 18S rRNA genes have been deposited in the NCBI SRA (Sequence Read Archive) database and are accessible in BioProject PRJNA1128422 (https://www.ncbi.nlm.nih.gov/sra/PRJNA1128422). Mitochondrial cytochrome *b* gene sequences have been deposited in GenBank under accession numbers PQ030722–PQ03080. All data files and scripts used for the analyses can be found here: https://github.com/willhr99/Cyprinella.git.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Venn Diagram showing relative size and overlap of ASVs obtained from water and fish samples. **Figure S2:** Network of water sample microbiota correlated with fish sample microbiota. **Figure S3:** Network of eukaryotic clades correlated with microbial ASVs. **Table S1:**. Core microbiota of 14 ASVs found across all sites and at least 50% of samples. **Table S2:** All unique 18S rRNA gene gut content classifications for *C. lutrensis*, *C. venusta*, *C. trichroistia* and *C. callistia*.